

B.Tech. Degree VII Semester Special Supplementary Examination
April 2016

ME 1702 VIBRATION AND NOISE CONTROL
(2012 scheme)

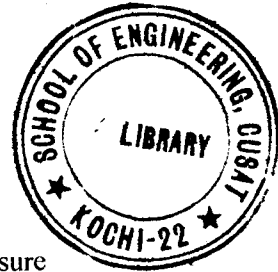
Time : 3 Hours

Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

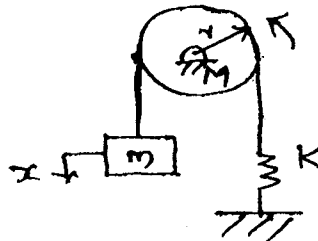
- I. (a) Derive an expression for logarithmic decrement.
(b) Define critical damping and damping ratio.
(c) Explain the concept of co-ordinate coupling.
(d) Write a note on accelerometer with a neat sketch.
(e) Discuss on the importance of Rayleigh's method in vibration analysis.
(f) What do you mean by whirling speed of a shaft?
(g) Explain the difference between sound intensity level and sound pressure level.
(h) What do you mean by annoyance of noise?



PART B

(4 × 15 = 60)

- II. (a) Determine the natural frequency of the spring mass pulley system shown in figure (8)

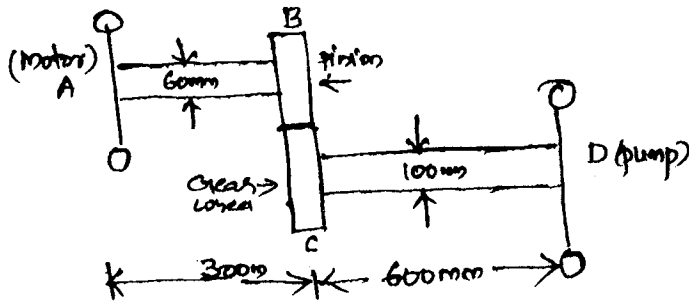


- (b) A vibratory system in a vehicle is to be designed with the following parameters: $k = 200 \text{ N/m}$, $c = 4 \text{ N/(m/s)}$, $m = 2 \text{ kg}$. Calculate the decrease of amplitude from its starting value after 4 complete oscillations and the frequency of oscillation. (7)

OR

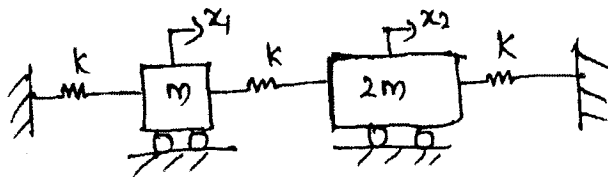
(P.T.O.)

- III. (a) An electric motor is supported on a spring and a dashpot. The spring has the stiffness of 6200 N/m and the dashpot offers resistance of 600 N at 5 m/s. The unbalanced mass 0.5 kg rotates at 5 cm radius and the total mass of vibratory system is 30 kg. The motor runs at 500 rpm. Determine (i) damping factor (ii) amplitude of vibration (iii) force exerted by the spring and dashpot on the motor. (8)
- (b) A torsional pendulum when immersed in oil indicates its natural frequency as 200 Hz. But when it was put to vibration in vacuum having no damping, its natural frequency was observed as 250 Hz. Find the value of damping factor of the oil. (7)
- IV. (a) What is a torsionally equivalent shaft? Explain. (5)
- (b) A motor drives a centrifugal pump through gearing, the pump speed being one third that of the motor. The shaft from the motor to the pinion is 60 mm diameter and 300 mm long. The moment of inertia of the motor is 500 kg m². The impeller shaft is 100 mm diameter and 600 mm long. The moment of inertia of the impeller is 1400 kg m². Neglecting the inertia of the gears and the shafts determine the frequency of torsional vibration of the system. The modulus of rigidity of the shaft is 80 GPa. (10)

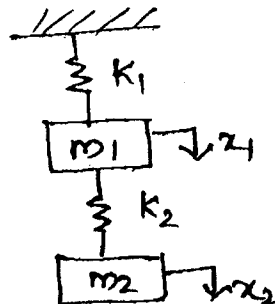


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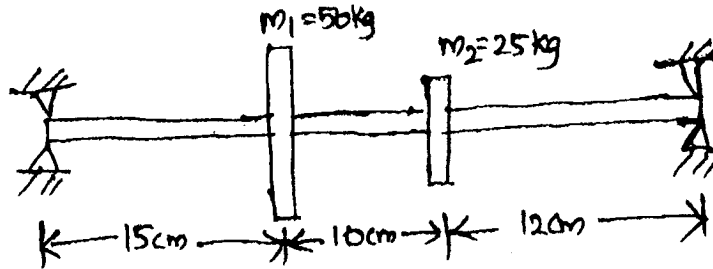
- V. (a) Derive the equation of motion for the system shown in figure by using Lagrange's equation. (7)



- (b) Determine the two natural frequencies of vibrations and the ratio of amplitudes of the motion m_1 and m_2 for the two modes of vibration of system shown in figure. (8)
- Given $m_1 = 3$ kg, $m_2 = 1$ kg, $k_1 = 30$ N/m and $k_2 = 50$ N/m



- VI. (a) Explain the Dunkerley's method in determining the frequency of a transversely vibrating system. (5)
- (b) Find the lowest natural frequency of transverse vibrations for the system shown in figure by Rayleigh's method. (10)



OR

- VII. (a) Derive suitable expression for longitudinal vibrations of a uniform beam fixed at one end and free at the other end. (11)
- (b) Explain the difference between continuous and discrete systems with suitable examples. (4)
- VIII. Explain the various methods of noise control, at the source, at the receiver and along the path. (15)

OR

- IX. (a) Explain the following. (10)
- (i) Noise (ii) Decibel (iii) Equivalent sound level (iv) Loudness
- (b) Explain the psychological effects of noise. (5)